English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE 12

Motion-Part-02

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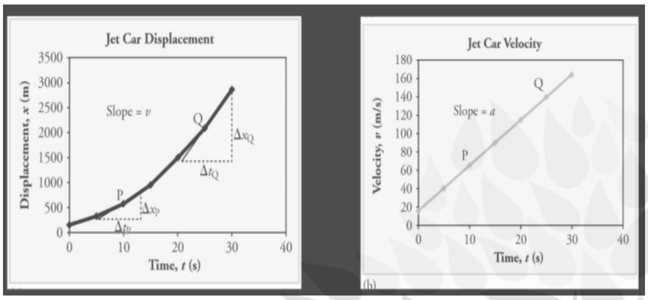
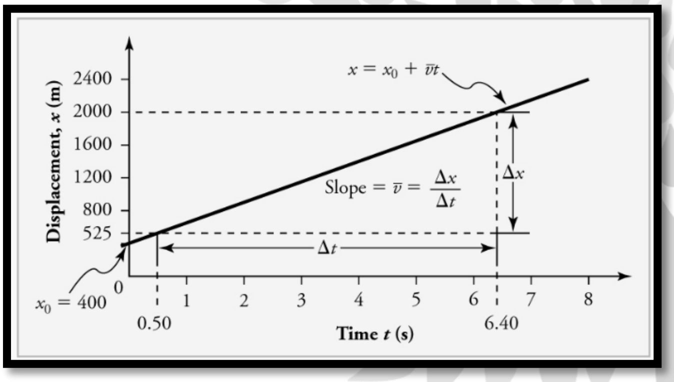
Graphical Analysis of Motion

Position-Time (s-t) Graph:

- The slope of the graph gives the velocity.

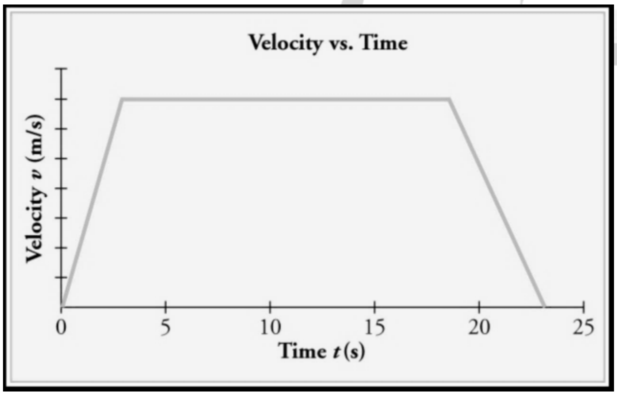


- A straight line indicates constant velocity



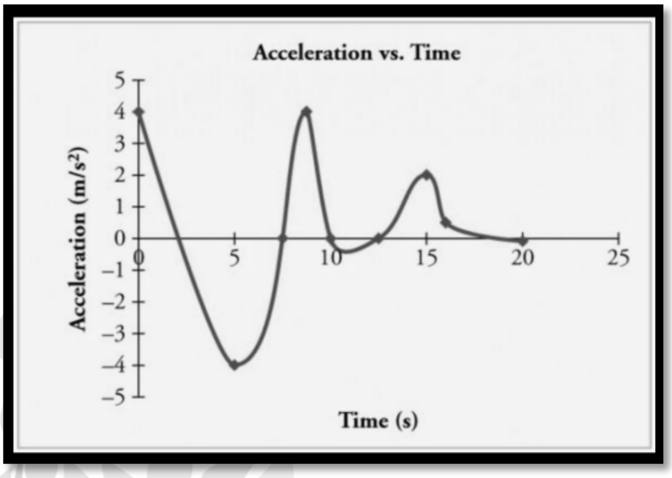
Velocity-Time (v-t) Graph:

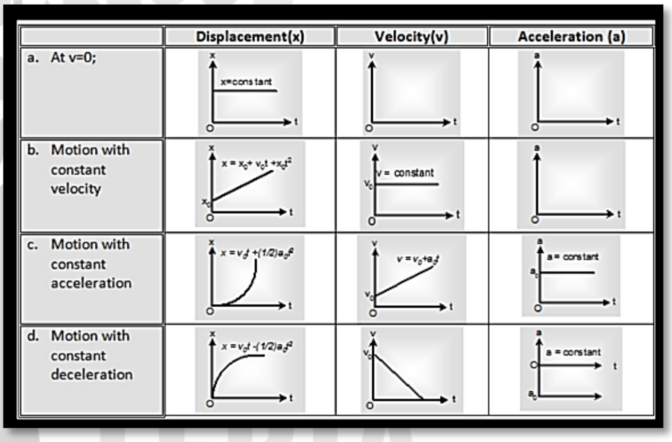
- The slope of the graph gives the acceleration.
- The area under the graph gives the displacement.
- A horizontal line indicates constant velocity (zero acceleration).



- **Acceleration-Time (a-t) Graph:**

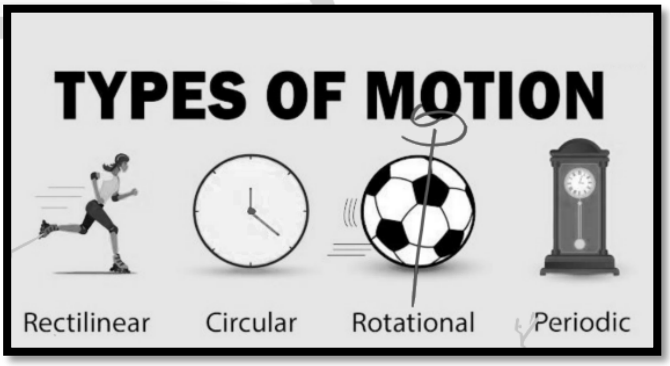
- The area under the graph gives the change in velocity.





Types of motion

(1) Translational motions: When all points of body move in some directing is same amount of time. It can be classified into two types.

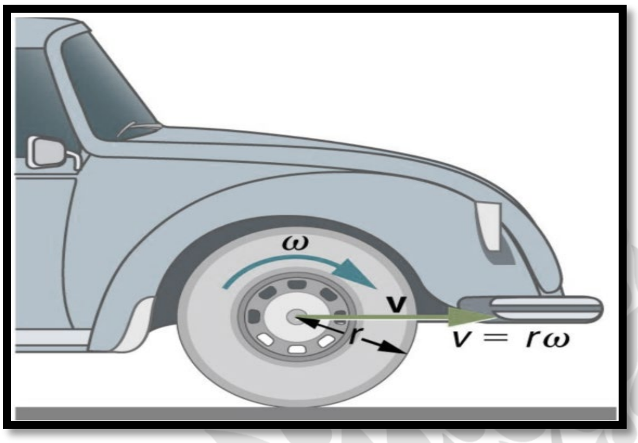


(i) Rectilinear motion : A person moving on straight called as rectilinear motion.

(ii) **Circular motion:** Moving along a curve path known as circular motion.

(2) **Rotational motion:** When a body spins around an axis inside.

E.g.- Rotation of earth on its own axis.



(3) **Periodic motion:** When motion repeats itself in fixed internal of time known as periodic motion.

E.g.- Beating of Heart, Earth's revolution around the sun.

(4) **Simple Hormonic motion (SHM) :** Back and forth repeated from known as SHM. Body moving to and fro along it's mean position.

Example : Pendulum motion.

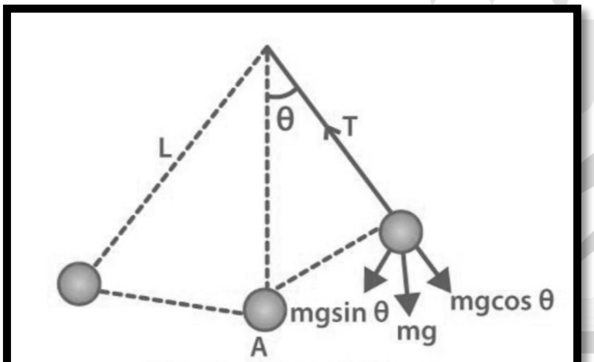
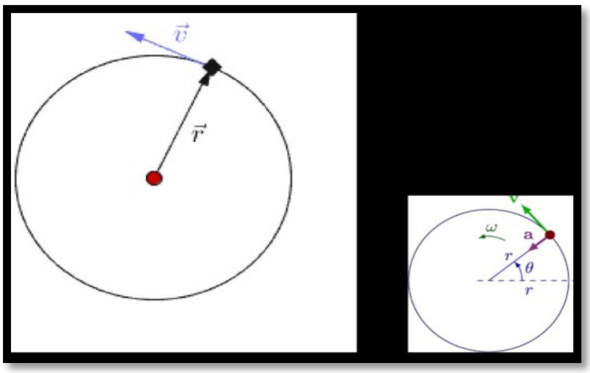


fig. simple pendulum



Newton's Laws of Motion

1. First Law of Motion (Law of Inertia)

Statement: A body at rest remains at rest, and a body in motion continues in uniform motion in a straight line unless acted upon by an external unbalanced force.

- Objects resist change in their state of motion.
- If no external force acts → velocity remains constant.

There are three laws given under Newton's law.

I. **(Low of Inertia/ Galileo's law of inertia**: A body continues its state of motion or rest until it is not compelled by ang external force.

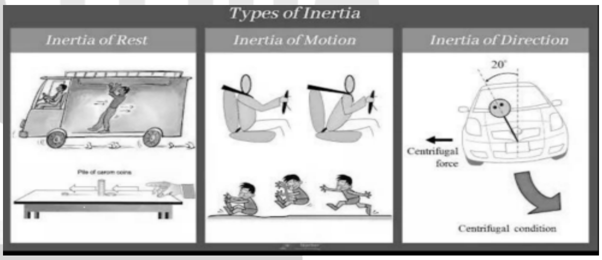
Law of inertia can be explained in three conditions: -

1. **Inertia of Rest:** when inertia apposes chang its state of rest.

   E,g- when bus suddenly street moving passengers fall backward.

State of motion.

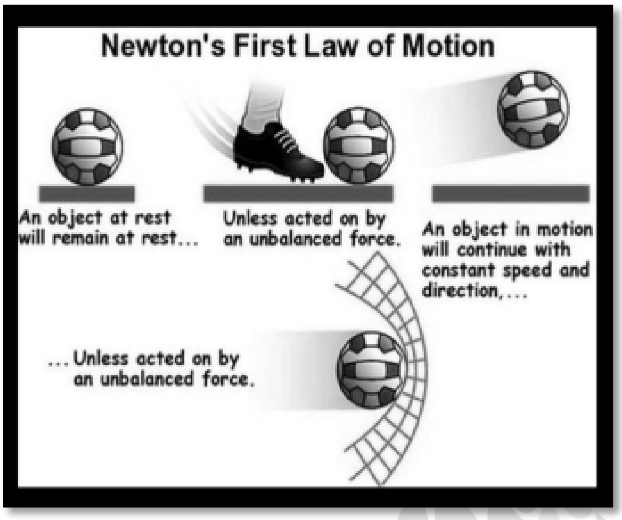
E.g.- When bus suddenly starts moving passengers fall backward.



2. **Inertia of motion:** Inertia opposes changes in its state of motion.

E.g.- Moving bus when sudden apply break passenger falls forward. A bowlder runs with the ball before throwing its so that velocity of bowler get added with the ball hence fast bowling occurs.

3. **Inertia of direction:** When inertia of body opposes changes in its direction.



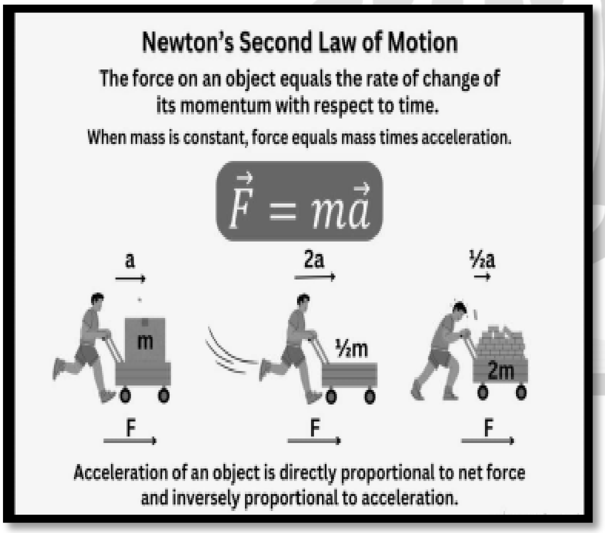
2. **Second Law of Motion (Law of Acceleration)**

Statement: The rate of change of momentum of an object is directly proportional to the applied force and takes place in the direction of force.

Formula: F = ma

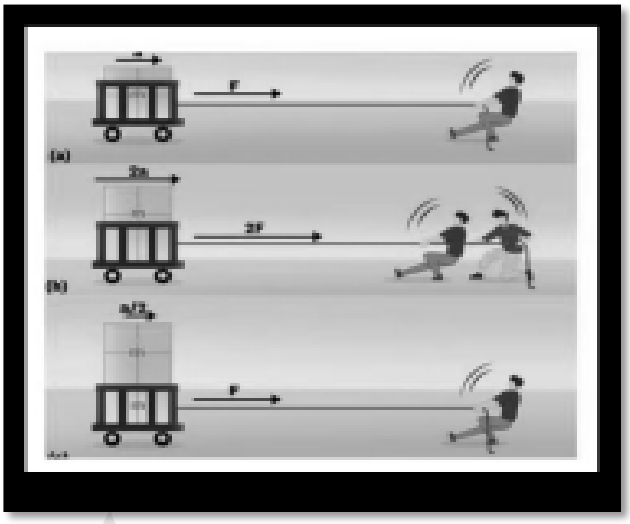
Where:

- F= Force (Newton, N)
- m= Mass (kg)
- a= Acceleration (m/s^2)



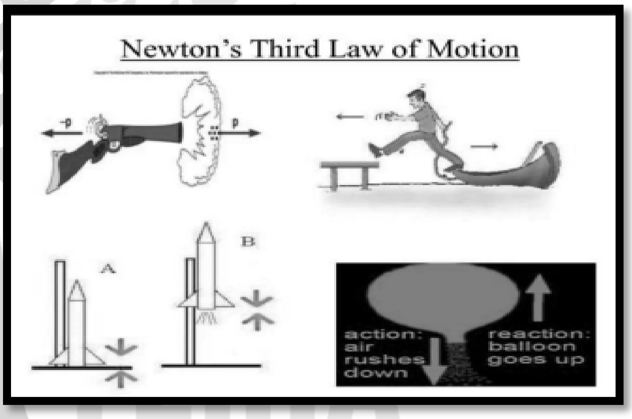
Second Law of Motion (Law of Acceleration)
Momentum: *p= mv*

- **If force applied increases** → acceleration increases proportionally.
- **Example:** Kicking a football harder increases its acceleration and speed.



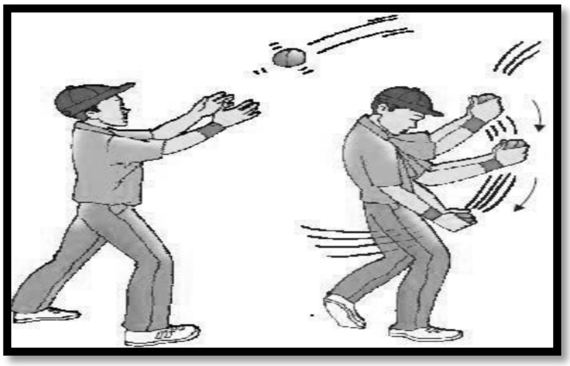
3. **Third Law of Motion (Action-Reaction Law)**

- **Statement:** For every action, there is an equal and opposite reaction.
- **Explanation:** Forces always occur in pairs but act on different bodies.



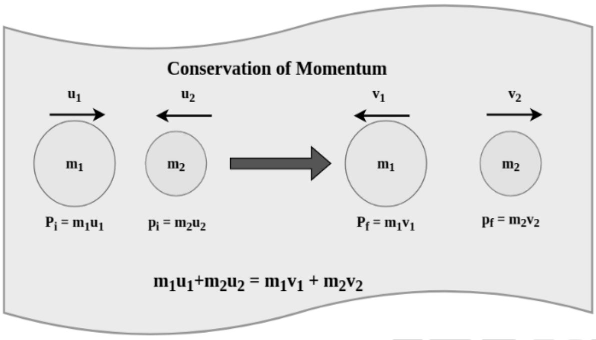
Examples:

- Rocket propulsion (gases pushed downward→ rocket moves upward).
- Rowing a boat (oar pushes water backward → boat moves forward).
- Walking (foot pushes ground backward → ground pushes foot forward).



Law of conservation of linear momentum

If net force applied on a given system becomes zero then initial momentum becomes its final momentum.

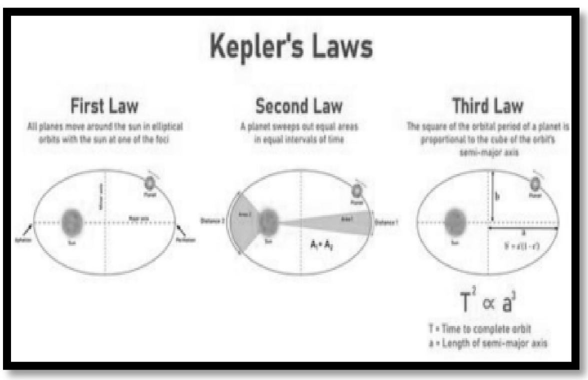


- $P_f = P_i$
- $mv_1 = mv_2$

Kepler's Law

Kepler's Laws of Planetary Motion describe how planets move around the Sun. They were formulated by Johannes Kepler in the early 17th century.

1. **Law of Orbits (Elliptical Law)**
   - Planets move in elliptical orbits around the Sun.
   - The Sun is at one focus of the ellipse (not at the center).
   - This explains why distance between planet and Sun keeps changing.
2. **Law of Areas (Equal Area Law)**
   - A line joining the planet and the Sun sweeps equal areas in equal time intervals.
   - When the planet is closer (perihelion) → moves faster
   - When it is farther (aphelion) → moves slower
   - Shows variation in orbital speed.



3. **Law of Periods (Harmonic Law)**
   - Square of orbital period (T^2) is proportional to cube of average distance (R^3) from the Sun.

$$T^2 \propto R^3$$

   - Farther planets take much longer time to revolve.
   - Example: Earth vs Mars → Mars takes more time.

Friction:

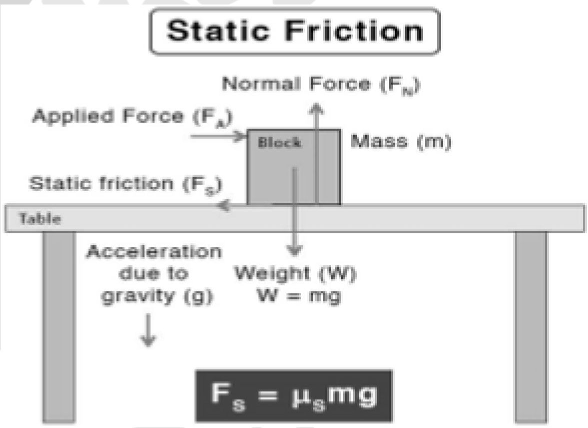
Friction is a force that opposes the relative motion (or tendency of motion) between two surfaces in contact. It acts along the surface, opposite to motion.

Types of Friction

I. Static Friction

- Acts when objects are at rest
- Prevents motion from starting

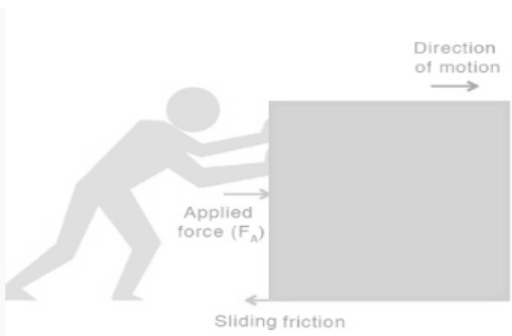
Maximum value: $f_s^{max} = \mu_s N$



II. Kinetic (Sliding) Friction

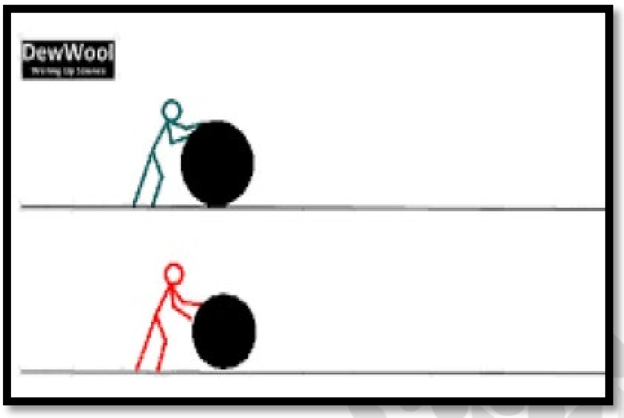
$f_s^{max} = \mu_s N$

- Acts when object is moving
- Always less than static friction



II. Rolling Friction

- Acts in rolling motion (wheels, balls)
- Smallest among all → why wheels are used



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